DATA ANALYTICS — PROJECT 2

Exploratory Data Analysis (EDA) Report

Prepared for: DecodeLabs Industrial Training Programme | Batch 2026

Dataset: Dataset for Data Analytics-2.xlsx

Coverage: 01 January 2023 to 30 June 2025

Executive Summary

This report examines 1,200 order records across 14 fields. The dataset contains 1,189 unique customer IDs and records a combined order value of 1,264,761.96. Average order value is 1,053.97, while the median is 823.62. The difference between mean and median, together with a positive skewness of 0.89, indicates that higher-value orders pull the average upward.

Printer has the highest order count (181), while Chair has the highest combined recorded order value (195,620.11). Eight TotalPrice observations exceed the IQR upper fence of 3,330.41. The main data-quality issue is 309 missing CouponCode values (25.8% of records).

Important interpretation: TotalPrice is treated in this report as recorded order value, not confirmed cash revenue. The dataset includes Cancelled and Returned orders, so financial totals should not be interpreted as realised revenue without a business rule for refunds, cancellations and fulfilment.

1. Project Objective and Method

The objective is to understand the dataset's structure, summarise its numerical and categorical variables, identify patterns over time, inspect unusual values, and examine relationships among numeric fields. The analysis uses descriptive statistics, frequency counts, monthly aggregation, the interquartile range (IQR) for outlier screening, and Pearson correlation for numeric associations.

2. Dataset Overview

Item	Result
Records	1,200
Columns	16
Date coverage	01 Jan 2023 – 30 Jun 2025
Unique customers	1,189
Duplicate rows	0
Duplicate OrderID values	0
Missing values	309 (all in CouponCode)
Total recorded order value	1,264,761.96

The fields describe order identifiers, dates, customers, products, quantities, unit prices, shipping addresses, payment methods, order statuses, tracking numbers, cart sizes, coupon codes, referral sources and calculated total prices.

3. Data Quality Checks

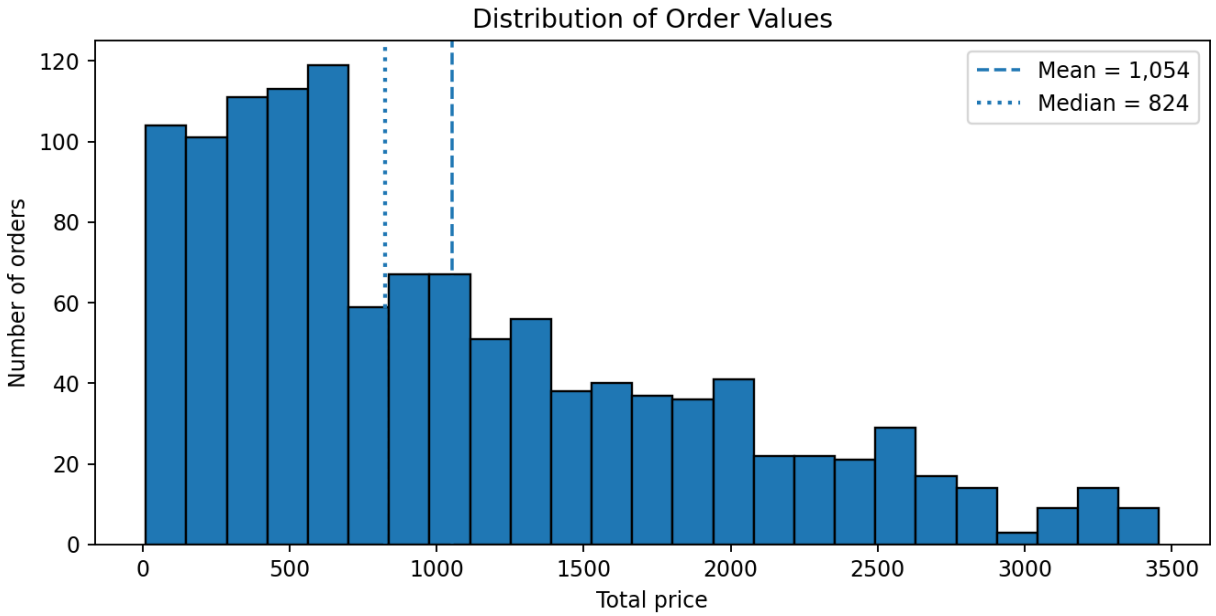
Check	Result	Interpretation
Missing values	309 CouponCode blanks	Only incomplete field; retain and disclose.
Duplicate rows	0	No exact duplicate rows detected.
Duplicate OrderIDs	0	Order identifier is unique in this file.
Date validity	1,200/1,200 valid	All dates parsed successfully.
TotalPrice arithmetic	0 mismatches	TotalPrice agrees with Quantity × UnitPrice to 0.01.

Treatment recommendation: do not automatically replace missing CouponCode values with a particular coupon. Use a label such as “Missing / not recorded” in analysis unless the data owner confirms that blank means no coupon was used. The missing values may represent either no coupon or unrecorded coupon information.

4. Descriptive Statistics

The table summarises the main numerical variables. Standard deviation measures spread around the mean; quartiles show the distribution's middle half.

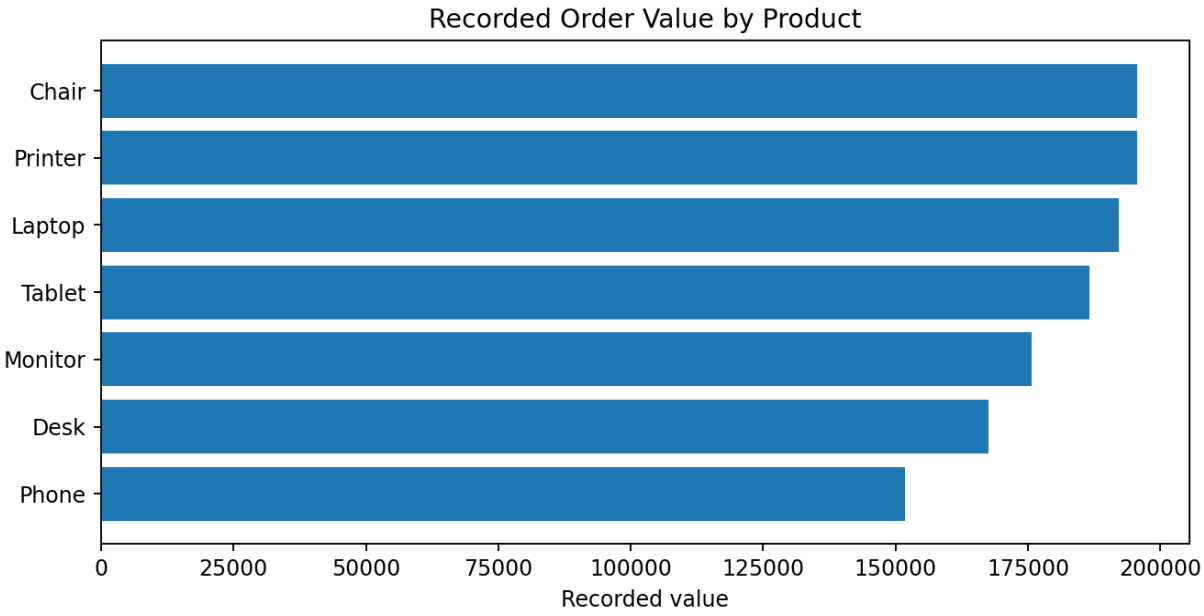
Variable	Count	Mean	Median	Std. Dev.	Min	Q1	Q3	Max
Quantity	1,200	2.95	3.00	1.41	1.00	2.00	4.00	5.00
UnitPrice	1,200	356.41	364.21	197.18	11.39	186.06	521.57	699.93
ItemsInCart	1,200	5.49	5.00	2.28	1.00	4.00	7.00	10.00
TotalPrice	1,200	1,053.97	823.62	819.86	11.39	410.52	1,578.47	3,456.40



Interpretation: average order value (1,053.97) is higher than the median (823.62). This is consistent with a right-skewed distribution: a relatively small number of large orders raise the mean. A typical order is therefore better represented by the median when describing the centre of this dataset.

5. Product Analysis

Product	Orders	Units	Recorded value	Average order value
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58



Printer generated the most orders (181), but Chair had the highest combined recorded value (195,620.11), narrowly ahead of Printer (195,612.61). Phone had the fewest orders (156) and the lowest combined recorded value (151,722.39).

6. Order Status and Customer Acquisition

Order status counts are relatively evenly distributed. This should be reviewed operationally because the file contains substantial volumes of Cancelled and Returned orders.

Order status	Orders	Share	Recorded value
Cancelled	250	20.8%	276,396.21
Returned	247	20.6%	243,277.70
Pending	237	19.8%	256,328.15
Shipped	235	19.6%	246,159.58
Delivered	231	19.2%	242,600.32

Payment method	Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Referral source	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

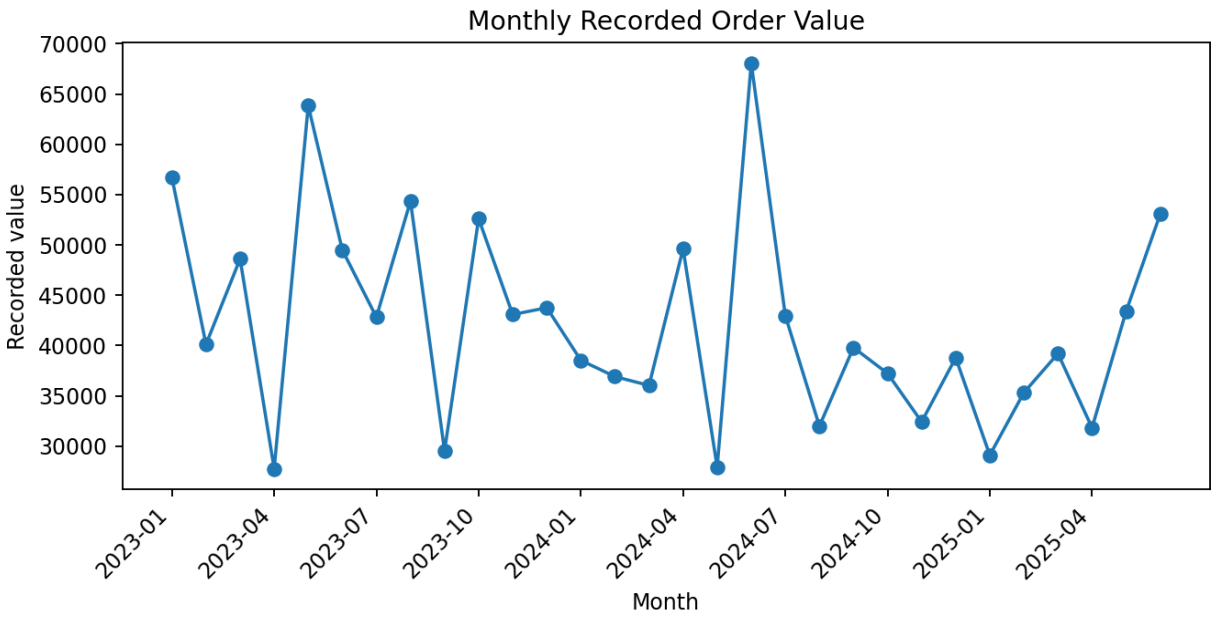
Payment methods and referral sources are also spread fairly evenly in terms of order counts. This dataset alone does not establish that a payment method or referral channel causes higher sales or better customer quality; further analysis of conversion, acquisition cost and customer value would be needed.

Coupon codes

FREESHIP appears on 313 records, WINTER15 on 292, SAVE10 on 286, and 309 records have no CouponCode recorded. Missing coupon values should remain a separate category until their meaning is confirmed.

7. Time Trend Analysis

The dataset covers January 2023 through June 2025. Monthly recorded order value was highest in 2024-06 (68,068.54) and lowest in 2023-04 (27,751.71).



Year-level totals are shown below. The 2025 period contains only January to June, so its total must not be compared directly with the full 12-month totals for 2023 and 2024.

Year	Period represented	Orders	Recorded value
2023	Jan–Dec	510	552,643.24
2024	Jan–Dec	459	480,235.87
2025	Jan–Jun only	231	231,882.85

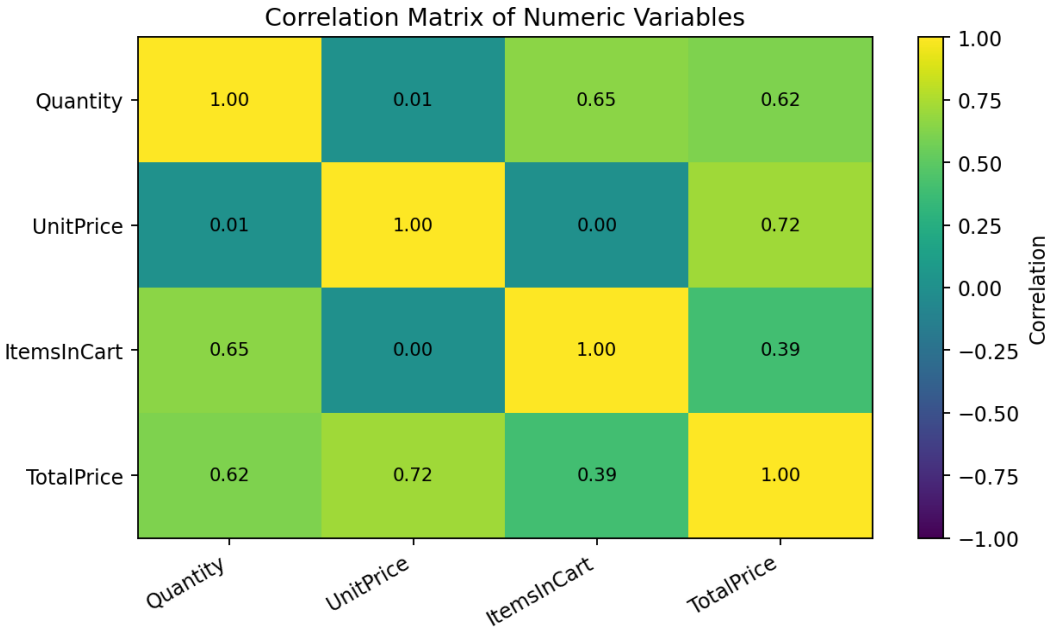
8. Outlier Analysis

Using the IQR method for TotalPrice: Q1 = 410.52, median = 823.62, Q3 = 1,578.47, and IQR = 1,167.95. The upper fence is Q3 + 1.5 × IQR = 3,330.41; the lower fence is -1,341.41. There are 8 orders above the upper fence and none below the lower fence.

Order ID	Date	Product	Qty	Unit price	Total price
ORD200789	2023-08-17	Tablet	5	691.28	3,456.40
ORD201122	2023-06-07	Monitor	5	678.19	3,390.95
ORD200632	2023-05-02	Laptop	5	678.16	3,390.80
ORD200469	2023-11-26	Chair	5	676.98	3,384.90
ORD200328	2023-02-28	Tablet	5	674.04	3,370.20
ORD200107	2023-03-27	Printer	5	670.75	3,353.75
ORD200326	2024-07-01	Laptop	5	670.48	3,352.40
ORD201065	2023-10-30	Printer	5	666.80	3,334.00

These are statistical flags, not proof of data errors. Since TotalPrice is calculated as Quantity × UnitPrice and all arithmetic checks pass, the high values may be legitimate large orders. They should be reviewed against business rules before removal or correction.

9. Correlation Analysis



TotalPrice has a positive correlation with UnitPrice (r = 0.717) and Quantity (r = 0.615). This is expected because TotalPrice is calculated from quantity multiplied by unit price. Quantity and ItemsInCart are also positively associated (r = 0.650). Correlation describes association and does not, by itself, prove causation.

10. Key Findings

1. The file contains 1,200 orders and 16 variables across the period 01 Jan 2023 to 30 Jun 2025.

2. Total recorded order value is 1,264,761.96; average order value is 1,053.97 and median order value is 823.62.

3. Printer has the highest order count (181); Chair has the highest combined recorded order value (195,620.11).

4. There are 8 IQR-flagged high order values. They are not automatically errors.

5. There are 309 missing CouponCode values, while no duplicate rows or OrderIDs were found and all TotalPrice calculations reconcile.

11. Recommendations

• Investigate the reasons for cancelled and returned orders, then track cancellation and return rates by product, payment method, referral source and month.

• Confirm the meaning of blank CouponCode values with the data owner before recoding them.

• Review the eight high-value orders flagged by IQR against order records; retain them if valid.

• Monitor monthly order value and order volume together. Use comparable full periods when making year-on-year comparisons.

• Extend the analysis with customer-level repeat-purchase behaviour, product-level return rates, and net revenue after cancellations/refunds if those fields become available.

12. Limitations

This dataset does not provide product cost, profit margin, discount amount, refund amount, delivery duration, customer demographics or a clear definition of whether TotalPrice for cancelled/returned orders represents realised revenue. The report therefore analyses recorded order values and observed patterns only. It does not estimate profit, customer lifetime value, or causal effects. The 2025 data covers six months only.

13. Conclusion

The exploratory analysis provides a structured view of order activity, product performance, recorded order values, order status, time patterns and relationships among numeric variables. The dataset is largely complete and internally consistent for the checked fields, with missing coupon information as the main quality issue. The findings offer a starting point for deeper operational analysis, particularly around cancellations, returns and customer acquisition.

Prepared as an Exploratory Data Analysis deliverable for DecodeLabs Project 2.